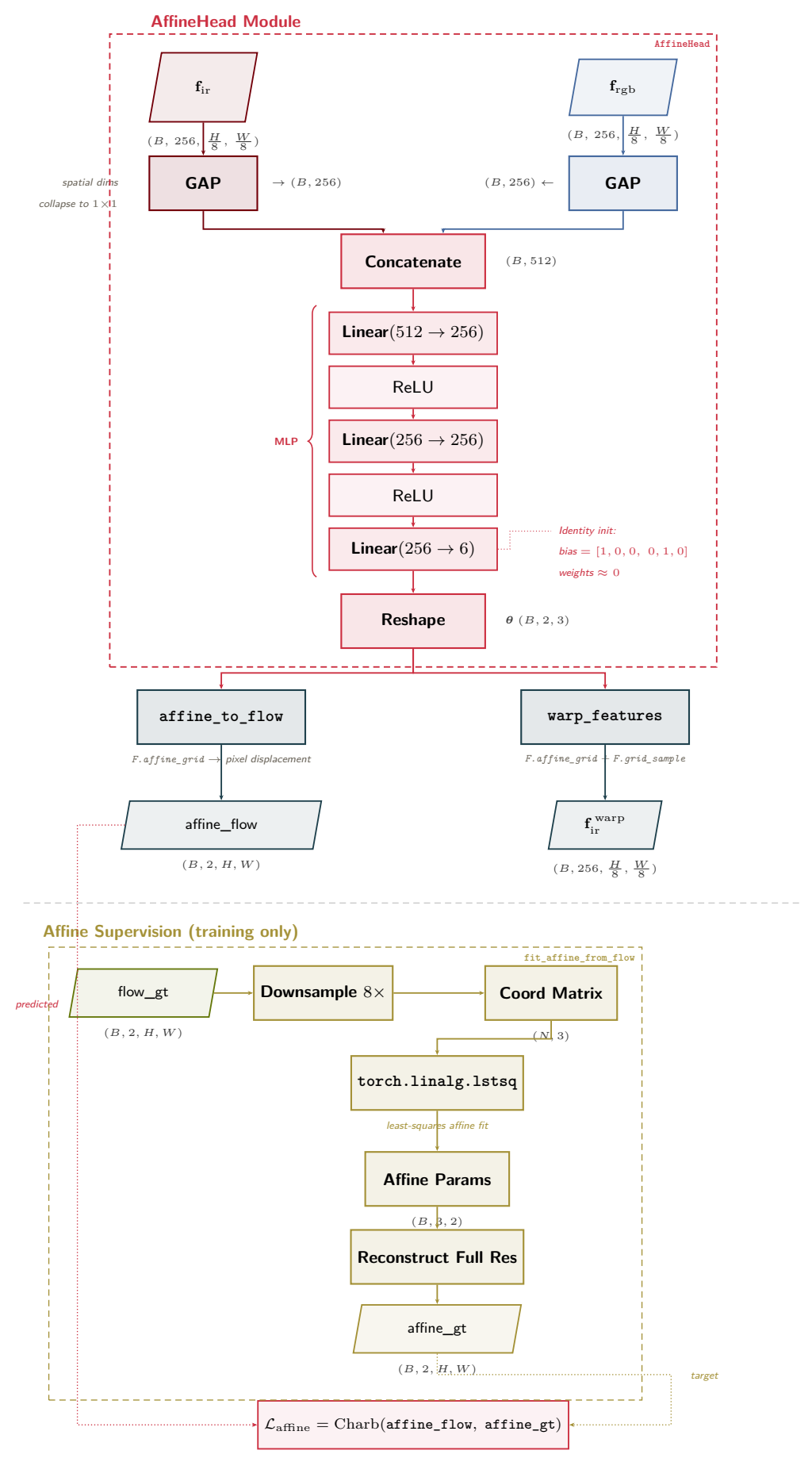
AffineHead Module
AffineHead
$\mathbf{f}_{\mathrm{ir}}$
$(B,\ 256,\ \frac{H}{8},\ \frac{W}{8})$
$\mathbf{f}_{\mathrm{rgb}}$
$(B,\ 256,\ \frac{H}{8},\ \frac{W}{8})$
spatial dims
collapse to 1×1
GAP
$\rightarrow (B, 256)$
$(B, 256) \leftarrow$
GAP
Concatenate
$(B, 512)$
MLP
Linear$(512 \rightarrow 256)$
ReLU
Linear$(256 \rightarrow 256)$
ReLU
Linear$(256 \rightarrow 6)$
Identity init:
bias $= [1, 0, 0,\ 0, 1, 0]$
weights ≈ 0
Reshape
$\boldsymbol{\theta}$ $(B, 2, 3)$
affine_to_flow
F.affine_grid → pixel displacement
warp_features
F.affine_grid + F.grid_sample
affine_flow
$(B, 2, H, W)$
$\mathbf{f}_{\mathrm{ir}}^{\mathrm{warp}}$
$(B, 256, \frac{H}{8}, \frac{W}{8})$
Affine Supervision (training only)
fit_affine_from_flow
predicted
flow_gt
$(B, 2, H, W)$
Downsample $8\times$
Coord Matrix
$(N, 3)$
torch.linalg.lstsq
least-squares affine fit
Affine Params
$(B, 3, 2)$
Reconstruct Full Res
affine_gt
$(B, 2, H, W)$
target
$\mathcal{L}_{\mathrm{affine}} = \mathrm{Charb}(\texttt{affine_flow},\ \texttt{affine_gt})$